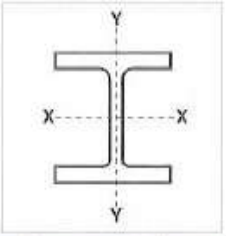


## 4a (i) Beams ( Imperial Units ) - Part I

| | Designation Size | | Unit Weight | | Height Of Beam | Width Of Beam | Thickness | | Root Radius | Depth Between Fillets | Area Of Section | Ratios For Local Buckling | | Dimensions For Detailing | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Flange | Web | | | | Flange | Web | End Clearance | Notch | |
| | | | | | H | B | T | t | r | d | A | b/T | d/t | C | N | n |
| | in | (mm) | lb/ft | kg/m | mm | mm | mm | mm | mm | mm | cm² | | | mm | mm | mm |
| W21 | 21 x 8 1/4 | (533 x 210) | 48 | 71.4 | 523.2 | 206.8 | 10.9 | 8.9 | 12.7 | 476.5 | 91 | 11.62 | 43.7 | 6 | 109 | 23 |
| | | | 55 | 81.8 | 528.3 | 208.8 | 13.2 | 9.6 | 12.7 | 476.5 | 105 | 7.91 | 49.6 | 7 | 110 | 26 |
| | | | 62 | 92.3 | 533.1 | 209.3 | 15.6 | 10.1 | 12.7 | 476.5 | 117 | 6.71 | 47.2 | 7 | 110 | 28 |
| | | | 68 | 101.2 | 536.7 | 210.0 | 17.4 | 10.8 | 12.7 | 476.5 | 129 | 6.03 | 44.1 | 7 | 110 | 30 |
| | | | 73 | 108.6 | 539.5 | 210.8 | 18.8 | 11.6 | 12.7 | 476.5 | 139 | 5.61 | 41.1 | 8 | 110 | 32 |
| | | | 82 | 122.0 | 544.5 | 211.9 | 21.3 | 12.7 | 12.7 | 476.5 | 155 | 4.97 | 37.5 | 8 | 110 | 34 |
| | | | 83 | 123.5 | 544.3 | 212.2 | 21.2 | 13.1 | 12.7 | 476.5 | 157 | 5.00 | 36.4 | 9 | 110 | 34 |
| | | | 93 | 138.4 | 549.1 | 213.9 | 23.6 | 14.7 | 12.7 | 476.5 | 176 | 4.53 | 32.3 | 9 | 110 | 36 |
| | 21 x 12 1/4 | (533 x 312) | 101 | 150.3 | 542.5 | 312.2 | 20.3 | 12.7 | 12.7 | 476.5 | 192 | 7.68 | 37.5 | 8 | 160 | 33 |
| | | | 111 | 165.2 | 546.4 | 313.4 | 22.2 | 14.0 | 12.7 | 476.6 | 211 | 7.05 | 34.1 | 9 | 160 | 35 |
| | | | 122 | 181.6 | 550.7 | 314.7 | 24.4 | 15.2 | 12.7 | 476.5 | 231 | 6.45 | 31.3 | 10 | 160 | 37 |
| | | | 132 | 196.4 | 554.5 | 316.0 | 26.3 | 16.5 | 12.7 | 476.5 | 250 | 6.01 | 28.9 | 10 | 160 | 39 |
| | | | 147 | 218.8 | 560.3 | 317.8 | 29.2 | 18.3 | 12.7 | 476.5 | 279 | 5.44 | 26.1 | 11 | 160 | 42 |
| | | | 166 | 247.0 | 571.0 | 315.5 | 34.5 | 19.1 | 12.7 | 476.5 | 315 | 4.57 | 25.0 | 12 | 158 | 47 |
| | | | 182 | 270.8 | 577.1 | 317.8 | 37.6 | 21.1 | 12.7 | 476.5 | 346 | 4.23 | 22.6 | 13 | 158 | 50 |
| | | | 201 | 299.1 | 585.0 | 319.0 | 41.4 | 23.1 | 12.7 | 476.8 | 382 | 3.85 | 20.6 | 14 | 158 | 54 |
| | | | 223 | 331.9 | 593.0 | 322.0 | 45.5 | 25.4 | 12.7 | 476.6 | 422 | 3.54 | 18.8 | 15 | 158 | 58 |
| | | | 248 | 369.1 | 603.0 | 324.0 | 50.5 | 26.2 | 12.7 | 476.6 | 460 | 3.21 | 18.2 | 15 | 159 | 63 |
| | | | 275 | 409.2 | 613.0 | 327.0 | 55.6 | 31.0 | 12.7 | 476.4 | 521 | 2.94 | 15.4 | 18 | 158 | 68 |
| W24 | 24 x 7 | (610 x 178) | 55 | 81.8 | 598.7 | 177.9 | 12.8 | 10.0 | 12.7 | 547.6 | 105 | 6.93 | 54.6 | 7 | 94 | 26 |
| | | | 61 | 90.8 | 602.5 | 178.4 | 15.0 | 10.6 | 12.7 | 547.1 | 116 | 5.95 | 51.6 | 7 | 94 | 28 |
| | | | 62 | 92.3 | 603.0 | 178.8 | 15.0 | 10.9 | 12.7 | 547.6 | 118 | 5.96 | 50.1 | 7 | 94 | 28 |
| | 24 x 9 | (610 x 229) | 68 | 101.2 | 602.6 | 227.6 | 14.8 | 10.5 | 12.7 | 547.6 | 129 | 7.69 | 52.2 | 7 | 119 | 28 |
| | | | 76 | 113.1 | 607.6 | 228.2 | 17.3 | 11.1 | 12.7 | 547.6 | 144 | 6.60 | 49.3 | 8 | 119 | 30 |
| | | | 84 | 125.0 | 612.2 | 229.0 | 19.3 | 11.9 | 12.7 | 548.2 | 158 | 5.93 | 46.1 | 8 | 119 | 32 |
| | | | 94 | 139.9 | 617.2 | 230.2 | 22.1 | 13.1 | 12.7 | 547.6 | 178 | 5.21 | 41.8 | 9 | 119 | 35 |
| | | | 103 | 153.3 | 623.1 | 228.6 | 24.9 | 14.0 | 12.7 | 547.9 | 195 | 4.59 | 39.1 | 9 | 117 | 38 |
| | | | 114 | 169.7 | 628.9 | 229.9 | 27.9 | 15.5 | 12.7 | 547.7 | 218 | 4.12 | 35.3 | 10 | 117 | 41 |
| | | | 128 | 190.5 | 635.0 | 231.5 | 30.9 | 17.0 | 12.7 | 547.8 | 242 | 3.75 | 32.2 | 11 | 117 | 44 |
| | | | 146 | 217.3 | 643.1 | 234.1 | 35.1 | 19.6 | 12.7 | 547.5 | 278 | 3.33 | 27.9 | 12 | 117 | 48 |
| | | | 163 | 242.6 | 651.0 | 235.9 | 39.1 | 21.6 | 12.7 | 547.4 | 310 | 3.02 | 25.3 | 13 | 117 | 52 |
| | | | 181 | 269.4 | 658.6 | 238.5 | 42.9 | 23.9 | 12.7 | 547.4 | 343 | 2.78 | 22.9 | 14 | 117 | 56 |
| | | | 198 | 294.7 | 667.0 | 240.5 | 47.0 | 25.9 | 12.7 | 547.6 | 376 | 2.56 | 21.1 | 15 | 117 | 60 |
| | | | 218 | 324.4 | 675.1 | 243.1 | 51.0 | 28.4 | 12.7 | 547.7 | 412 | 2.38 | 19.3 | 16 | 117 | 64 |
| | | | 239 | 355.7 | 685.0 | 245.6 | 55.9 | 31.0 | 12.7 | 547.8 | 454 | 2.20 | 17.7 | 18 | 117 | 69 |
| | 24 x 12 (UB) | (610 x 305) | 100 | 148.8 | 612.4 | 304.8 | 19.7 | 11.8 | 16.5 | 540.0 | 190 | 7.74 | 45.8 | 8 | 157 | 36 |
| | | | 120 | 178.6 | 620.2 | 307.1 | 23.6 | 14.1 | 16.5 | 540.0 | 228 | 6.51 | 38.3 | 9 | 157 | 40 |
| | | | 160 | 238.1 | 635.8 | 311.4 | 31.4 | 18.4 | 16.5 | 540.0 | 303 | 4.96 | 29.3 | 11 | 157 | 48 |
| | 24 x 12 3/4 | (610 x 324) | 104 | 154.8 | 611.1 | 323.8 | 19.1 | 12.7 | 12.7 | 547.6 | 198 | 8.50 | 43.1 | 8 | 166 | 32 |
| | | | 117 | 174.1 | 616.2 | 325.1 | 21.6 | 14.0 | 12.7 | 547.6 | 222 | 7.53 | 39.2 | 9 | 166 | 34 |
| | | | 131 | 194.9 | 621.8 | 326.5 | 24.4 | 15.4 | 12.7 | 547.6 | 249 | 6.70 | 35.6 | 10 | 166 | 37 |
| | | | 146 | 217.3 | 628.4 | 327.7 | 27.7 | 16.5 | 12.7 | 547.6 | 277 | 5.92 | 33.2 | 10 | 166 | 40 |
| | | | 162 | 241.1 | 635.0 | 329.1 | 31.0 | 17.9 | 12.7 | 547.6 | 308 | 5.31 | 30.6 | 11 | 166 | 44 |
| | | | 176 | 261.9 | 641.1 | 327.4 | 34.0 | 19.1 | 12.7 | 547.6 | 333 | 4.81 | 28.7 | 12 | 164 | 47 |
| | | | 192 | 285.7 | 646.9 | 328.9 | 37.1 | 20.6 | 12.7 | 547.3 | 363 | 4.44 | 26.6 | 12 | 164 | 50 |
| | | | 207 | 308.0 | 653.0 | 330.5 | 39.9 | 22.1 | 12.7 | 547.8 | 392 | 4.14 | 24.8 | 13 | 164 | 53 |
| | | | 229 | 340.8 | 661.0 | 333.0 | 43.9 | 24.4 | 16.5 | 540.2 | 435 | 3.79 | 22.1 | 14 | 164 | 60 |
| | | | 250 | 372.0 | 669.0 | 335.0 | 48.0 | 26.4 | 16.5 | 540.0 | 475 | 3.49 | 20.5 | 15 | 164 | 65 |
| | | | 279 | 415.2 | 679.0 | 338.0 | 53.1 | 29.5 | 16.5 | 539.8 | 530 | 3.18 | 18.3 | 17 | 164 | 70 |
| | | | 306 | 455.4 | 689.0 | 340.0 | 57.9 | 32.0 | 16.5 | 540.2 | 579 | 2.94 | 16.9 | 18 | 164 | 74 |
| | | | 335 | 498.5 | 699.0 | 343.0 | 63.0 | 35.1 | 13.0 | 547.0 | 635 | 2.72 | 15.6 | 20 | 164 | 76 |
| | | | 370 | 550.6 | 711.0 | 347.0 | 69.1 | 38.6 | 13.0 | 546.8 | 702 | 2.51 | 14.2 | 21 | 164 | 82 |

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.

## Dimensions and Properties




# 4a (i) Beams ( Imperial Units ) - Part II

| Unit Weight | | Surface Area per Metre | Second Moment Of Area | | Radius Of Gyration | | Elastic Modulus | | Plastic Modulus | | Buckling Parameter | Torsional Index | Warping Constant | Torsional Constant | Designation Size | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Axis x-x | Axis y-y | Axis x-x | Axis y-y | Axis x-x | Axis y-y | Axis x-x | Axis y-y | | | | | | |
| lb/ft | kg/m | $m^2$ | $cm^4$ | $cm^4$ | cm | cm | $cm^3$ | $cm^3$ | $cm^3$ | $cm^3$ | u | x | $dm^6$ | $cm^4$ | in | |
| 48 | 71.43 | 1.85 | 39917 | 1611 | 20.9 | 4.20 | 1524 | 156 | 1753 | 244 | 0.858 | 47.9 | 1.1 | 33.4 | 21 x 8 1/4 | W21 |
| 55 | 81.85 | 1.85 | 47540 | 2008 | 21.3 | 4.38 | 1800 | 192 | 2059 | 300 | 0.864 | 41.6 | 1.33 | 51.5 | | |
| 62 | 92.27 | 1.86 | 55230 | 2389 | 21.7 | 4.51 | 2072 | 228 | 2360 | 356 | 0.872 | 36.5 | 1.60 | 75.7 | | |
| 68 | 101.19 | 1.87 | 61520 | 2692 | 21.9 | 4.57 | 2292 | 256 | 2612 | 399 | 0.874 | 33.2 | 1.82 | 101 | | |
| 73 | 108.64 | 1.88 | 66820 | 2943 | 21.9 | 4.60 | 2477 | 279 | 2828 | 436 | 0.875 | 30.9 | 1.99 | 126 | | |
| 82 | 122.03 | 1.89 | 76040 | 3388 | 22.1 | 4.67 | 2793 | 320 | 3196 | 500 | 0.877 | 27.6 | 2.32 | 178 | | |
| 83 | 123.52 | 1.89 | 76240 | 3389 | 22.0 | 4.65 | 2801 | 319 | 3212 | 500 | 0.874 | 27.6 | 2.32 | 181 | | |
| 93 | 138.40 | 1.90 | 86170 | 3868 | 22.1 | 4.68 | 3139 | 362 | 3617 | 569 | 0.873 | 24.9 | 2.67 | 251 | | |
| 101 | 150.30 | 2.29 | 100800 | 10320 | 22.9 | 7.33 | 3715 | 661 | 4147 | 1012 | 0.885 | 27.8 | 7.03 | 217 | 21 x 12 1/4 | |
| 111 | 165.19 | 2.30 | 111300 | 11410 | 23.0 | 7.36 | 4074 | 728 | 4565 | 1117 | 0.885 | 25.6 | 7.84 | 284 | | |
| 122 | 181.56 | 2.31 | 123300 | 12680 | 23.1 | 7.40 | 4476 | 806 | 5032 | 1238 | 0.885 | 23.4 | 8.78 | 373 | | |
| 132 | 196.44 | 2.32 | 134200 | 13850 | 23.2 | 7.44 | 4842 | 876 | 5462 | 1348 | 0.884 | 21.8 | 9.66 | 469 | | |
| 147 | 218.76 | 2.33 | 151200 | 15650 | 23.3 | 7.49 | 5396 | 985 | 6116 | 1519 | 0.884 | 19.8 | 11.0 | 643 | | |
| 166 | 247.03 | 2.34 | 178000 | 18110 | 23.8 | 7.58 | 6233 | 1148 | 7080 | 1766 | 0.891 | 17.2 | 13.0 | 983 | | |
| 182 | 270.85 | 2.36 | 197200 | 20150 | 23.9 | 7.63 | 6834 | 1268 | 7807 | 1956 | 0.890 | 15.9 | 14.7 | 1278 | | |
| 201 | 299.12 | 2.38 | 220700 | 22450 | 24.1 | 7.67 | 7547 | 1408 | 8670 | 2175 | 0.890 | 14.6 | 16.6 | 1700 | | |
| 223 | 331.86 | 2.40 | 247700 | 25390 | 24.2 | 7.76 | 8355 | 1577 | 9656 | 2442 | 0.890 | 13.4 | 19.0 | 2263 | | |
| 248 | 369.06 | 2.43 | 278900 | 28710 | 24.6 | 7.90 | 9250 | 1772 | 10730 | 2739 | 0.894 | 12.3 | 21.9 | 2996 | | |
| 275 | 409.24 | 2.45 | 316900 | 32530 | 24.7 | 7.91 | 10340 | 1990 | 12120 | 3096 | 0.890 | 11.2 | 25.3 | 4131 | | |
| 55 | 81.85 | 1.87 | 56020 | 1210 | 23.2 | 3.40 | 1871 | 136 | 2200 | 219 | 0.843 | 48.3 | 1.04 | 49.2 | 24 x 7 | W24 |
| 61 | 90.78 | 1.88 | 63880 | 1426 | 23.5 | 3.51 | 2120 | 160 | 2480 | 256 | 0.850 | 43.2 | 1.23 | 68.6 | | |
| 62 | 92.27 | 1.88 | 64580 | 1436 | 23.4 | 3.49 | 2142 | 161 | 2512 | 258 | 0.848 | 42.8 | 1.24 | 71.0 | | |
| 68 | 101.19 | 2.07 | 75780 | 2915 | 24.2 | 4.76 | 2515 | 256 | 2881 | 400 | 0.864 | 43.1 | 2.52 | 77.0 | 24 x 9 | |
| 76 | 113.10 | 2.08 | 87320 | 3434 | 24.6 | 4.88 | 2874 | 301 | 3281 | 469 | 0.870 | 38.0 | 2.99 | 111 | | |
| 84 | 125.01 | 2.09 | 97540 | 3872 | 24.8 | 4.95 | 3187 | 338 | 3639 | 528 | 0.872 | 34.5 | 3.40 | 149 | | |
| 94 | 139.89 | 2.11 | 111800 | 4505 | 25.0 | 5.03 | 3622 | 391 | 4142 | 611 | 0.875 | 30.6 | 3.99 | 216 | | |
| 103 | 153.28 | 2.11 | 125000 | 4972 | 25.3 | 5.04 | 4012 | 435 | 4595 | 680 | 0.878 | 27.6 | 4.45 | 295 | | |
| 114 | 169.65 | 2.12 | 141400 | 5670 | 25.4 | 5.09 | 4495 | 493 | 5167 | 773 | 0.878 | 24.8 | 5.12 | 411 | | |
| 128 | 190.48 | 2.14 | 158400 | 6415 | 25.6 | 5.15 | 4990 | 554 | 5757 | 871 | 0.879 | 22.6 | 5.85 | 554 | | |
| 146 | 217.27 | 2.16 | 183900 | 7544 | 25.7 | 5.21 | 5718 | 644 | 6643 | 1019 | 0.877 | 20.0 | 6.97 | 819 | | |
| 163 | 242.57 | 2.18 | 207900 | 8606 | 25.9 | 5.27 | 6386 | 730 | 7455 | 1157 | 0.878 | 18.2 | 8.06 | 1123 | | |
| 181 | 269.36 | 2.20 | 232800 | 9768 | 26.1 | 5.34 | 7069 | 819 | 8299 | 1304 | 0.877 | 16.7 | 9.26 | 1495 | | |
| 198 | 294.66 | 2.22 | 259400 | 10980 | 26.3 | 5.41 | 7778 | 913 | 9173 | 1458 | 0.878 | 15.4 | 10.6 | 1952 | | |
| 218 | 324.42 | 2.24 | 287700 | 12330 | 26.4 | 5.47 | 8522 | 1014 | 10110 | 1625 | 0.877 | 14.3 | 12.0 | 2517 | | |
| 239 | 355.67 | 2.27 | 322200 | 13950 | 26.6 | 5.55 | 9406 | 1136 | 11220 | 1826 | 0.877 | 13.2 | 13.8 | 3309 | | |
| 100 | 148.82 | 2.39 | 125900 | 9308 | 25.7 | 7.00 | 4111 | 611 | 4594 | 937 | 0.886 | 32.7 | 8.17 | 200 | 24 x 12 | |
| 120 | 178.58 | 2.41 | 153000 | 11410 | 25.9 | 7.07 | 4935 | 743 | 5547 | 1144 | 0.886 | 27.7 | 10.2 | 340 | (UB) | |
| 160 | 238.11 | 2.45 | 209500 | 15840 | 26.3 | 7.23 | 6589 | 1017 | 7486 | 1574 | 0.886 | 21.3 | 14.5 | 785 | | |
| 104 | 154.77 | 2.47 | 129200 | 10790 | 25.6 | 7.39 | 4227 | 666 | 4734 | 1023 | 0.880 | 33.6 | 9.46 | 196 | 24 x 12 3/4 | |
| 117 | 174.11 | 2.48 | 147200 | 12380 | 25.8 | 7.47 | 4776 | 762 | 5360 | 1170 | 0.882 | 30.0 | 10.9 | 280 | | |
| 131 | 194.95 | 2.50 | 167300 | 14160 | 25.9 | 7.55 | 5383 | 867 | 6057 | 1335 | 0.883 | 26.8 | 12.6 | 395 | | |
| 146 | 217.27 | 2.51 | 190800 | 16260 | 26.2 | 7.66 | 6074 | 993 | 6845 | 1527 | 0.886 | 23.9 | 14.7 | 559 | | |
| 162 | 241.08 | 2.53 | 215400 | 18440 | 26.4 | 7.74 | 6784 | 1121 | 7670 | 1726 | 0.888 | 21.6 | 16.8 | 770 | | |
| 176 | 261.92 | 2.53 | 236600 | 19950 | 26.6 | 7.73 | 7380 | 1218 | 8369 | 1878 | 0.890 | 19.9 | 18.4 | 995 | | |
| 192 | 285.73 | 2.55 | 260400 | 22030 | 26.8 | 7.79 | 8050 | 1340 | 9163 | 2068 | 0.890 | 18.4 | 20.5 | 1280 | | |
| 207 | 308.05 | 2.56 | 283900 | 24050 | 26.9 | 7.84 | 8695 | 1455 | 9936 | 2250 | 0.890 | 17.2 | 22.6 | 1593 | | |
| 229 | 340.79 | 2.58 | 319000 | 27090 | 27.1 | 7.90 | 9652 | 1627 | 11090 | 2523 | 0.890 | 15.7 | 25.8 | 2162 | | |
| 250 | 372.04 | 2.60 | 353900 | 30170 | 27.3 | 7.97 | 10580 | 1801 | 12220 | 2797 | 0.890 | 14.5 | 29.1 | 2809 | | |
| 279 | 415.20 | 2.62 | 400500 | 34310 | 27.5 | 8.04 | 11800 | 2030 | 13720 | 3162 | 0.889 | 13.2 | 33.6 | 3824 | | |
| 306 | 455.38 | 2.65 | 445200 | 38100 | 27.7 | 8.11 | 12920 | 2241 | 15120 | 3498 | 0.889 | 12.2 | 37.9 | 4940 | | |
| 335 | 498.53 | 2.68 | 494700 | 42580 | 27.9 | 8.19 | 14150 | 2483 | 16670 | 3885 | 0.889 | 11.3 | 42.9 | 6420 | | |
| 370 | 550.62 | 2.71 | 557510 | 48400 | 28.2 | 8.30 | 15682 | 2790 | 18599 | 4377 | 0.889 | 10.4 | 49.6 | 8525 | | |

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.